Insegnamento di Identificazione, Stima e Controllo Robusto
Modulo di *Controllo Robusto*

Norms for signals and systems

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Norms of vectors

The norm $N(x)$ of a vector $x \in \mathbb{R}^n$ is a function $N: \mathbb{R}^n \to \mathbb{R}^+$ satisfying the following properties:

- ✓ $N(x) \geq 0$;
- ✓ $N(x) = 0 \iff x = 0$;
- ✓ $N(x + y) \leq N(x) + N(y)$;
- ✓ $N(\alpha x) \leq |\alpha| N(x), \forall \alpha \in \mathbb{R}$;

Denoting by $|x_i|$ the absolute value of x_i, the ℓ_2-norm of vector x is defined by:

$$\|x\|_2 = \sqrt{x^T x} = \left(\sum_{i=1}^{n} |x_i|^2 \right)^{\frac{1}{2}}$$

The ℓ_2-norm is a special case of a more general family of norms, called ℓ_p-norm where $p \in \mathbb{N}^+$.

Norms of vectors

The ℓ_p-norm of a vector x is defined by:

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{\frac{1}{p}}, \forall\ 1 \leq p \leq \infty$$

The ℓ_∞-norm of a vector x is defined by:

$$\|x\|_\infty = \max_i |x_i| = \lim_{p\to\infty} \|x\|_p$$

Norms of signals

L_2-norm: For a scalar-valued signal $v(t)$ defined for $t \geq 0$ the L_2-norm is defined as

$$\|v\|_2 = \sqrt{\int_0^\infty v^2(\tau)\mathrm{d}\tau}$$

A physical interpretation of the L_2-norm is that if $v(t)$ is a voltage or a current, then $\|v\|_2^2$ is proportional to the total *energy* associated with the signal.

We can define the L_2-norm for the Laplace-transformed signal $V(s)$ on the imaginary axis according to

$$\|V\|_2 = \sqrt{\frac{1}{2\pi}\int_{-\infty}^{\infty} |V(j\omega)|^2 \mathrm{d}\omega}$$

By *Parseval's theorem*, the time domain and frequency domain L_2-norm are equal, i.e. $\|v\|_2 = \|V\|_2$.

The factor $^1/_{2\pi}$ is introduced to get consistency with the 2-norm of the corresponding impulse response.

Norms of signals

For a vector-valued signal $v(t) = [v_1(t), \dots, v_m(t)]^T$, the L_2-norm generalizes in a natural way to

$$\|v\|_2 = \sqrt{\int_0^\infty \sum_{i=1}^m v_i^2(\tau)\mathrm{d}\tau} = \sqrt{\int_0^\infty v(\tau)^T v(\tau)\mathrm{d}\tau}$$

representing its energy. Similarly in the Laplace-domain, signal $\mathrm{V}(s) = [V_1(s), \dots, V_m(s)]^T$, by the Parseval's theorem, we have

$$\|V\|_2 = \sqrt{\frac{1}{2\pi j}\oint_{C^-} V^T(-s)V(s)\mathrm{d}s} \underbrace{=}_{s=j\omega} \sqrt{\frac{1}{2\pi}\int_{-\infty}^{\infty} V^T(-j\omega)V(j\omega)\mathrm{d}\omega} =$$

$$= \sqrt{\frac{1}{2\pi}\int_{-\infty}^{\infty} \sum_{i=1}^m V_i(-j\omega)V_i(j\omega)\mathrm{d}\omega} = \sqrt{\frac{1}{2\pi}\int_{-\infty}^{\infty} \sum_{i=1}^m |V_i(j\omega)|^2\mathrm{d}\omega}$$

C^- is a contour following the imaginary axis of the complex plane and closed through a semi-circle with infinite radius in the left half plane.

Norms of signals

The *residue theorem*, from complex analysis, states as follows: let f be be analytic inside and on a simple closed contour C, except at finitely many isolated singularities $\lambda_1, \lambda_2, \dots, \lambda_n$ inside C. Then:

$$\oint_{C^-} f(s)\mathrm{d}s = 2\pi j \sum_{k=1}^{n} \mathrm{Res}(f, \lambda_k)$$

Consequently, we have

$$\|V\|_2 = \sqrt{\sum_{\text{poles of } V(s)} \mathrm{Res}(V^T(-s)V(s), \lambda_k)}$$

Let λ_i be a pole (with negative real part) of $V(s)$ with multiplicity n_i, residue $\mathrm{Res}(V^T(-s)V(s), \lambda_i)$ is defined as follows:

$$\mathrm{Res}(V^T(-s)V(s), \lambda_k) = \frac{1}{(n_i-1)!}\frac{d^{n_i-1}}{ds^{n_i-1}}(s-\lambda_i)^{n_i}V^T(-s)V(s)\Big|_{s=\lambda_i}$$

Norms of signals

Example 1: Let $x(t)$ be defined as follows where T is a positive constant:

$$x(t) = e^{-\frac{t}{T}}, \qquad \forall t \geq 0$$

The L_2-norm of signal $x(t)$ is:

$$\|x(t)\|_2 = \sqrt{\int_0^\infty x^T(\tau)x(\tau)\mathrm{d}\tau} = \sqrt{\int_0^\infty \mathrm{e}^{-\frac{2\tau}{T}}\mathrm{d}\tau} = \sqrt{\frac{T}{2}e^{-\frac{2t}{T}}\Big|_0^\infty} = \sqrt{\frac{T}{2}}$$

It can be equivalently computed using the Laplace transform $X(s)$ of $x(t)$:

$$X(s) = \frac{1}{s+\frac{1}{T}} = \frac{T}{Ts+1} \quad \Longrightarrow X^T(-s)X(s) = \frac{T^2}{(1+Ts)(1-Ts)}$$

Assuming $T > 0$, the unique pole $\lambda_1 = -\frac{1}{T}$ has negative part and its multiplicity $n_1 = 1$.

Norms of signals

Then, $\mathrm{Res}(X^T(-s)X(s), \lambda_1)$ is computed as follows

$$\mathrm{Res}(X^T(-s)X(s), \lambda_1) = \frac{1}{(1-1)!}\frac{d^{1-1}}{ds^{1-1}}(s-\lambda_1)^1 X^T(-s)X(s)\Big|_{s=-\frac{1}{\tau}}$$

$$= \left(s+\frac{1}{T}\right)\frac{T^2}{(1+Ts)(1-Ts)}\Bigg|_{s=-\frac{1}{T}} = \frac{T}{(1-sT)}\Bigg|_{s=-\frac{1}{T}} = \frac{T}{2}$$

and finally, $\|x(t)\|_2 = \sqrt{\frac{T}{2}}$.

Norms of signals

Example 2: Let now $x(t)$ be defined as follows

$$x(t) = \begin{bmatrix} e^{-t} \\ e^{-2t} \end{bmatrix}, \qquad \forall t \geq 0$$

The L_2-norm of signal $x(t)$ is:

$$\|x(t)\|_2 = \sqrt{\int_0^\infty x^T(\tau)x(\tau)\mathrm{d}\tau} = \sqrt{\int_0^\infty e^{-2\tau} + e^{-4\tau}\mathrm{d}\tau}$$

$$= \sqrt{\frac{e^{-2t}}{-2} + \frac{e^{-4t}}{-4}\Bigg|_0^\infty} = \sqrt{\frac{1}{2} + \frac{1}{4}} = \sqrt{\frac{3}{4}}$$

The Laplace transform $X(s)$ of $x(t)$:

$$X(s) = \begin{bmatrix} \frac{1}{s+1} & \frac{1}{s+2} \end{bmatrix}^T \Rightarrow X^T(-s)X(s)$$

$$= \frac{1}{(-s+1)(s+1)} + \frac{1}{(-s+2)(s+2)}$$

Norms of signals

$X(s)$ has two poles with negative real part, namely $\lambda_1 = -1$ and $\lambda_2 = -2$, each with multiplicity equal to 1. We obtain

$$\mathrm{Res}(X^T(-s)X(s), \lambda_1) = \frac{1}{(1-1)!}\frac{d^{1-1}}{ds^{1-1}}(s-\lambda_1)^1 X^T(-s)X(s)\Big|_{s=-1}$$

$$= (s+1)\left(\frac{1}{(-s+1)(s+1)} + \frac{1}{(-s+2)(s+2)}\right)\Bigg|_{s=-1} = \frac{1}{2}$$

while,

$$\mathrm{Res}(X^T(-s)X(s), \lambda_2) = \frac{1}{(1-1)!}\frac{d^{1-1}}{ds^{1-1}}(s-\lambda_2)^1 X^T(-s)X(s)\Big|_{s=-2}$$

$$= (s+2)\left(\frac{1}{(-s+1)(s+1)} + \frac{1}{(-s+2)(s+2)}\right)\Bigg|_{s=-2} = \frac{1}{4}$$

And hence $\|x(t)\|_2 = \sqrt{\frac{3}{4}}$.

Norms of signals

$\boldsymbol{L_p}$ **norm**: the L_2-norm is a special case of the L_p-norm, defined as

$$\|v\|_p = \sqrt[p]{\int_0^\infty |v(\tau)|^p \mathrm{d}\tau}$$

In particular, the L_1-norm is the integral of the absolute value,

$$\|v\|_1 = \sqrt{\int_0^\infty |v(\tau)| \, \mathrm{d}\tau}$$

The L_p-norm can be defined in the Laplace domain, but for $p \neq 2$, no relationship corresponding to the Parseval's theorem exists between the time-domain and the s-domain norms.

Remark: the notation L in L_p refers to the fact that the integrand should be *Lebesgue*-integrable for the integral to exist. This is a generalization of the standard (Riemann) integral to a more general class of functions.

Norms of signals

***L_∞*-norm**: as $p \to \infty$, the L_p-norm tends to the so called ∞-norm, or L_∞-norm, which can be characterized as

$$\|v\|_\infty = \max_t |v(t)|$$

supposing that the maximum exists.
However, there is in general no guarantee that the maximum exists, and therefore the correct way is to define the L_∞-norm as the least upper bound (or *supremum*) of the absolute value

$$\|v\|_\infty = \sup_t |v(t)|$$

The L_∞-norm of a vector signal v is defined by

$$\|v\|_\infty = \sup_{t \in \mathbb{R}^+} \max_i |V_i(t)|$$

Norms of signals

Remark. The symbol ∞ comes from

$$\max_t |v(t)| = \lim_{p\to\infty}\left(\int_{-\infty}^{\infty}|v(\tau)|^p \mathrm{d}\tau\right)^{\frac{1}{p}}$$

Example: The function

$$f(x) = \begin{cases} x, & \text{if } |x| < 1 \\ 0, & \text{if } |x| \geq 1 \end{cases}$$

has no maximum. For any $x = x_1,\ x_1 = 1 - \epsilon$, there exists another value $x_2 = 1 - {}^{\epsilon}/_{2}$ such that $f(x_2) > f(x_1)$.

The least upper bound or supremum is defined as the least number M which satisfies $M \geq f(x), \forall x$, i.e.,

$$\sup_x f(x) = \min\{M \ : \ f(x) \geq M,\ \forall x\}$$

For the example $\sup_x f(x) = 1$.

Norms of signals

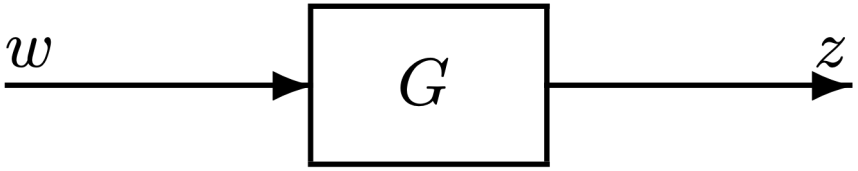
Examples: for the function $x(t) = e^{-t}$ defined above, we have

$$\|x(t)\|_\infty = \sup_{t\in\mathbb{R}^+} |e^{-t}| = 1$$

For the vector of signals $x(t) = \begin{bmatrix} e^{-t} \\ e^{-2t} \end{bmatrix}$ defined above

$$\|x(t)\|_\infty = \sup_{t\in\mathbb{R}^+} \max_i |x_i(t)| = \sup_{t\in\mathbb{R}^+} \max_i \{e^{-t}, e^{-2t}\} = \sup_{t\in\mathbb{R}^+} e^{-t} = 1$$

System norms



Given information about the allowed input signals $w(t)$, how large can the outputs $z(t)$ become?

For evaluating the output signal, we will use the 2-norm,

$$\|z(t)\|_2 = \sqrt{\sum_i \int_{-\infty}^{\infty} |z_i(\tau)|^2 d\tau}$$

and consider three inputs:

1) $w(t)$ is a series of unit impulses $\Longrightarrow$ $\boldsymbol{\mathcal{H}_2}$ **norm;**

2) $w(t)$ is a signal satisfying $\|w(t)\|_2 = 1 \Longrightarrow$ $\boldsymbol{\mathcal{H}_\infty}$ **norm;**

3) $w(t)$ is a signal satisfying $\|w(t)\|_2 = 1$, but $w(t) = 0$ for $t \geq 0$, and we only measure $z(t)$ for $t \geq 0 \Longrightarrow$ **Hankel norm.**

$\mathcal{H}_2$: The Hardy 2-space

The Hardy 2-space $\mathcal{H}_2$ consists of functions of a complex variable that are analytic in the open right-half of the complex plane and such that the norm

$$\|f\|_2 = \left\{ \sup_{\alpha>0} \frac{1}{2\pi} \int_{-\infty}^{\infty} f^*(\alpha + j\omega) f(\alpha + j\omega) d\omega \right\}^{\frac{1}{2}}$$

is finite. That is,

$$\mathcal{H}_2 = \{f : f(s) \text{ is analytic in } R_e(s) > 0 \text{ and } \|f\|_2 < \infty\}.$$

For any $f \in \mathcal{H}_2$, the boundary function defined as $f_b(j\omega) = \lim_{\alpha \to 0} f(\alpha + j\omega)$ exists for almost all ω (Fatou's theorem). The mapping $f \longmapsto f_b$ is linear and injective. Moreover, the superior value of f always happens for $\alpha = 0$. Therefore,

$$\|f\|_2^2 = \|f_b\|_2^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} f_b^*(j\omega) f_b(j\omega) d\omega$$

$\mathcal{H}_2$: The Hardy 2-space

Concluding, $\mathcal{H}_2$ stands for the Hardy 2-space of transfer function $f(s)$ with **bounded 2-norm** which is the set of **stable** and **strictly proper transfer functions**.

$\mathcal{H}_2$ norms of stable plants

Let $G(s)$ be a *strictly proper* transfer matrix of a stable LTV system. Let $g(t)$ be the impulse response of the system and $\mathcal{L}[g(t)]$ the Laplace transform of $g(t)$.
We know that $\mathcal{L}[g(t)] = G(s)$ and $g(t) = \mathcal{L}^{-1}[G(s)]$.

The $\mathcal{H}_2$ norm of transfer matrix $G(s)$ is defined in the **time domain** through the impulse response $g(t)$ of the system and represents the ***impulse response energy***.

For SISO or SIMO systems the definition of $\mathcal{H}_2$ norm comes directly from the one of L_2-norm of scalar-valued signal or vector-valued signal, respectively.

$\mathcal{H}_2$ norms of stable plants

For MIMO systems, let's write $g(t)$ by column as follows, where $g_i(t)$ represents system's response to the unit impulse $\delta(t)$ to input i:

$$g(t) = [g_1(t) \quad g_2(t) \quad \cdots \quad g_m(t)] = [g_{ij}(t)]$$

Then we have

$$\|G(s)\|_2 = \sqrt{\sum_{i=1}^{m} \int_0^\infty g_i^T(\tau) g_i(\tau) \mathrm{d}\tau} = \sqrt{\sum_{i,j}^{m} \int_0^\infty |g_{ij}(\tau)|^2 \mathrm{d}\tau}$$

$$= \sqrt{\int_0^\infty \mathrm{tr}(g^T(\tau) g(\tau)) \mathrm{d}\tau} = \sqrt{\int_0^\infty \mathrm{tr}\big(g(\tau) g^T(\tau)\big) \mathrm{d}\tau}$$

Notice that $\mathrm{tr}[A^T A] = \sum_{i,j} a_{ij}^2 = a_{11}^2 + \cdots + a_{nm}^2 = \|A\|_F^2$.

$\mathcal{H}_2$ norms of stable plants

From the preceding definition, the $\mathcal{H}_2$ norm of $G(s)$ is the L_2-norm of output signal $g_i(t)$ when a unit impulse $\delta(t)$ is successively applied to each input channels, one after another (allowing the output to settle to zero before applying an impulse to the next input).

Thus, if $G(s)$ is a matrix with transfer functions $G_{ij}(s)$ at row i and column j, then:

$$G(s) = \left[G_{ij}(s)\right] \Longrightarrow \|G(s)\|_2 = \sqrt{\sum_{i,j} \left\|G_{ij}(s)\right\|_2^2}$$

$\mathcal{H}_2$ norms of stable plants

The $\mathcal{H}_2$ norm of a transfer matrix $G(s)$ can be equivalently defined in the frequency domain by using the Parseval's Theorem.

Let C^- be a contour following the imaginary axis of the complex plane and closed through a semi-circle with infinite radius in the left half plane. By Parseval's theorem we get:

$$\|G(s)\|_2 = \sqrt{\int_0^\infty \mathrm{tr}(g^T(\tau)g(\tau))\mathrm{d}\tau} = \sqrt{\frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{tr}(G^T(-j\omega)G(j\omega))\mathrm{d}\omega}$$

$$= \sqrt{\frac{1}{2\pi j}\oint_{C^-} \mathrm{tr}(G^T(-s)G(s))\mathrm{d}s}$$

which represent the total area under the power spectral density, i.e. the **spectral power integral**. Similarly by replacing $G^T(-s)G(s)$ and $G^T(-j\omega)G(j\omega)$ with $G(s)G^T(-s)$ $G(j\omega)G^T(-j\omega)$.

$\mathcal{H}_2$ norms of stable plants

For example, for SISO systems $Z(s) = G(s)V(s)$ with $V(s) = 1$, we have

$$\|Z\|_2 = \sqrt{\frac{1}{2\pi}\int_{-\infty}^{\infty} |G(j\omega)|^2 \mathrm{d}\omega} = \|G\|_2$$

Consequently, the $\mathcal{H}_2$ norm can be also interpreted as **an average system gain taken over all frequencies.**

Remark 1: Note that $G(s)$ and $g(t)$ are dynamic *systems* while $G(j\omega)$ and $g(\tau)$ are constant *matrices* (for a value of ω or τ).

Remark 2: as soon as $G(s)$ is strictly proper and all its poles have negative real part its $\mathcal{H}_2$ norm exists.

Remark 3: the **spectral power integral** and the ***impulse response energy*** *are identical by the Parseval's theorem.*

$\mathcal{H}_2$ norms of stable plants

From the definition of the Frobenius norm $\|G(s)\|_F^2 = \mathrm{tr}(G^T(-s)G(s))$, we can also write

$$\|G(s)\|_2 = \sqrt{\frac{1}{2\pi}\int_{-\infty}^{\infty}\|G(j\omega)\|_F^2 \mathrm{d}\omega} = \sqrt{\frac{1}{2\pi j}\oint_{C^-}\|G(s)\|_F^2 \mathrm{d}s}$$

and, by using $\mathrm{tr}(G^T(-s)G(s)) = \sum_{i=1}^{q}\sigma_i^2(G(s))$, the $\mathcal{H}_2$ norm of a transfer matrix $G(s)$ can also be computed by using its singular values:

$$\|G(s)\|_2 = \sqrt{\frac{1}{2\pi j}\sum_{i=1}^{q}\oint_{C^-}\sigma_i^2\big(G(s)\big)\mathrm{d}s} = \sqrt{\frac{1}{2\pi}\sum_{i=1}^{q}\int_{-\infty}^{\infty}\sigma_i^2(G(j\omega))\mathrm{d}\omega}$$

$\mathcal{H}_2$ norms of stable plants

We can also use of the *residue theorem* to compute the $\mathcal{H}_2$ norm:

$$\|G(s)\|_2 = \sqrt{\frac{1}{2\pi j}\oint_{C^-} \mathrm{tr}(G^T(-s)G(s))\mathrm{d}s}$$

$$= \sqrt{\sum_{\text{Poles of } G(s)} \mathrm{Res}\big(\mathrm{tr}\big(G^T(-s)G(s)\big), \lambda_i\big)}$$

where, if λ_i is a pole of $G(s)$ with multiplicity n_i:

$$\mathrm{Res}\big(\mathrm{tr}(G^T(s)G(s), \lambda_i)\big)$$
$$= \frac{1}{(n_i-1)!}\frac{d^{n_i-1}}{ds^{n_i-1}}(s-\lambda_i)^{n_i}\left(\mathrm{tr}\big(G^T(-s)G(s)\big)\right)\Bigg|_{s=\lambda_i}$$

and, similarly by replacing $G^T(-s)G(s)$ with $G(s)G^T(-s)$ under the trace operation.

$\mathcal{H}_2$ norms

The $\mathcal{H}_2$ norm can also have a *stochastic interpretation*: the squared $\mathcal{H}_2$ norm represents the **steady-state variance of the output signal** when each component of the input is a stochastic white noise with unit covariance, i.e., $E[u(t)] = 0$, $E[u(t)u(t-\tau)] = \delta(\tau)I$.

Let $z(t)$ be the output of a Linear Time Invariant SISO system, g(t) its impulsive response and $G(s)$ its Laplace transform. The output signal to a white noise is given by convolution

$$z(t) = \int_0^\infty g(\tau)u(t-\tau)\mathrm{d}\tau$$

Consequently,

$$E[z^2(t)] = E\left[\int_0^\infty g(\tau_1)u(t-\tau_1)\mathrm{d}\tau_1 \cdot \int_0^\infty g(\tau_2)u(t-\tau_2)\mathrm{d}\tau_2\right]$$

$\mathcal{H}_2$ norms

By moving inside the integral the expectation, we obtain

$$E[z^2(t)] = \int_0^\infty \int_0^\infty g(\tau_1)g(\tau_2)E[u(t-\tau_2)u(t-\tau_1)]\,\mathrm{d}\tau_1\mathrm{d}\tau_2$$

And by using the white noise property $E[u(t-\tau_2)u(t-\tau_1)] = \delta(\tau_1-\tau_2)$, we get

$$E[z^2(t)] = \int_0^\infty \int_0^\infty g(\tau_1)g(\tau_2)\delta(\tau_1-\tau_2)\,\mathrm{d}\tau_1\mathrm{d}\tau_2$$

The Dirac delta collapses the double integral into a single integral ($\tau_1 \equiv \tau_2 \equiv \tau$):

$$E[z^2(t)] = \int_0^\infty g^2(\tau)\,\mathrm{d}\tau = \|G(s)\|_2^2$$

$\mathcal{H}_2$ norms

Concluding, hereafter the three faces of the $\mathcal{H}_2$ norm:

$$\|G\|_2^2 = \underbrace{\int_0^\infty \mathrm{tr}\left[g^T(t)g(t)\right] dt}_{\substack{\textbf{Face 1} \\ \text{impulse response energy} \\ \delta(t) \text{ input}}} \overset{\text{Parseval}}{=} \underbrace{\frac{1}{2\pi}\int_{-\infty}^\infty \mathrm{tr}\left[G^T(j\omega)\,G(j\omega)\right] d\omega}_{\substack{\textbf{Face 2} \\ \text{spectral power integral} \\ \text{all frequencies}}} \overset{\text{stationarity}}{=} \underbrace{\mathbb{E}\left[\|z(t)\|^2\right]}_{\substack{\textbf{Face 3} \\ \text{output variance at steady state} \\ \text{white noise input}}}$$

$\mathcal{H}_2$ norms

<u>Examples</u>: We wish to calculate the $\mathcal{H}_2$-norm of a second order system whose input-output relation reads:

$$\frac{1}{\omega_n^2}\ddot{y}(t) + \frac{2\zeta}{\omega_n}\dot{y}(t) + y(t) = Ku(t)$$

where $\zeta \geq 0$ is the damping ratio, $\omega_n \geq 0$ the natural frequency and K the static gain. We will assume in the following that the damping ratio ζ is lower than 1 such that the 2 poles λ_1 and λ_2 of the transfer function $F(s)$ are complex conjugates:

$$\begin{cases} \lambda_1 = -\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2} \\ \lambda_2 = -\zeta\omega_n - j\omega_n\sqrt{1-\zeta^2} \end{cases}$$

Consequently, t.f. $F(s)$ reads

$$F(s) = \frac{K\omega_n^2}{(s-\lambda_1)(s-\lambda_2)}$$

$\mathcal{H}_2$ norms

Rather than computing the $\mathcal{H}_2$-norm thanks to the impulse response $g(t) = \mathcal{L}^{-1}[F(s)]$ we will use the residue theorem. Indeed, the transfer function $F(s)$ is strictly proper and all its poles have negative real part thus its $\mathcal{H}_2$-norm exists and the residue theorem can be applied.

This leads to the following expression:

$$\|F(s)\|_2 = \sqrt{\sum_{\text{Poles of } F(s)} \operatorname{Res}(F^T(-s)F(s), \lambda_i)}$$

where

$$\operatorname{Res}(F^T(s)F(s), \lambda_i) = \frac{1}{(n_i - 1)!} \frac{d^{n_i - 1}}{ds^{n_i - 1}} (s - \lambda_i)^{n_i} F^T(-s)F(s) \Big|_{s=\lambda_i}$$

$\mathcal{H}_2$ norms

The multiplicity of pole λ_1 is $n_1 = 1$. We get:

$$\mathrm{Res}(F^T(s)F(s), \lambda_1) = (s-\lambda_1)F^T(-s)F(s)\Big|_{s=\lambda_1}$$

$$= (s-\lambda_1)\frac{K\omega_n^2}{(-s-\lambda_1)(-s-\lambda_2)}\frac{K\omega_n^2}{(s-\lambda_1)(s-\lambda_2)}\Bigg|_{s=\lambda_1}$$

$$= \frac{(K\omega_n^2)^2}{2\lambda_1(\lambda_1+\lambda_2)(\lambda_1-\lambda_2)}$$

Similarly, the multiplicity of pole λ_2 is $n_2 = 1$. We get:

$$\mathrm{Res}(F^T(s)F(s), \lambda_2) = (s-\lambda_2)F^T(-s)F(s)\Big|_{s=\lambda_2}$$

$$= (s-\lambda_2)\frac{K\omega_n^2}{(-s-\lambda_1)(-s-\lambda_2)}\frac{K\omega_n^2}{(s-\lambda_1)(s-\lambda_2)}\Bigg|_{s=\lambda_2}$$

$$= \frac{(K\omega_n^2)^2}{2\lambda_2(\lambda_2+\lambda_1)(\lambda_2-\lambda_1)}$$

$\mathcal{H}_2$ norms

Using the fact that λ_2 is the complex conjugate of λ_1, $\lambda_2 = \lambda_1^*$, we get:

$$\mathrm{Res}(F^T(-s)F(s), \lambda_2) = \left(\mathrm{Res}(F^T(-s)F(s), \lambda_1)\right)^*$$

Denoting by Re the real part of a complex number we get:

$$\|F(s)\|_2^2 = \sum_{\text{Poles of } F(s)} \mathrm{Res}(F^T(-s)F(s), \lambda_i)$$

$$= 2Re\left(\sum_{\text{Poles of } F(s)} \mathrm{Res}(F^T(-s)F(s), \lambda_1)\right)$$

$$= Re\left(\frac{(K\omega_n^2)^2}{\lambda_1(\lambda_1+\lambda_2)(\lambda_1-\lambda_2)}\right)$$

$\mathcal{H}_2$ norms

Let's multiply the numerator and the denominator by $\lambda_2 = \lambda_1^*$ such that $\lambda_1\lambda_2 = |\lambda_1|^2 = \omega_n^2$:

$$\|F(s)\|_2^2 = Re\left(\frac{(K\omega_n^2)^2\lambda_2}{\lambda_2\lambda_1(\lambda_1+\lambda_2)(\lambda_1-\lambda_2)}\right) =$$

$$= Re\left(\frac{(K\omega_n^2)^2\left(-\zeta\omega_n - j\omega_n\sqrt{1-\zeta^2}\right)}{\omega_n^2(2\zeta\omega_n)(2j\omega_n\sqrt{1-\zeta^2})}\right) =$$

$$= Re\left(\frac{K^2\left(-j\zeta\omega_n - \omega_n\sqrt{1-\zeta^2}\right)}{-2\zeta(2\sqrt{1-\zeta^2})}\right) = \frac{K^2}{4}\frac{\omega_n}{\zeta}$$

and hence,

$$\|F(s)\|_2 = \frac{K}{2}\sqrt{\frac{\omega_n}{\zeta}}$$

$\mathcal{H}_2$ norm and Gramians

Rather than directly evaluating the integral

$$\|G\|_2 = \sqrt{\frac{1}{2\pi}\int_{-\infty}^{\infty} |G(j\omega)|^2 \mathrm{d}\omega} \qquad \text{(SISO)}$$

$$\|G\|_2^2 = \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{tr}[G(-j\omega)^T G(j\omega)] d\omega \qquad \text{(MIMO)}$$

the $\mathcal{H}_2$ norm of a transfer function can be conveniently calculated in time-domain, starting from its state-space representation. Assume that the stable transfer function $G(s) = C(sI - A)^{-1}B$, with state-space representation

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$z(t) = Cx(t)$$

Integrating previous state-space equations from 0 to t,

$$z(t) = Ce^{At}x(0) + \int_0^t H(t-\tau)u(\tau)\mathrm{d}\tau \qquad H(\tau) = \begin{cases} Ce^{A\tau}B, & \text{if } \tau \geq 0 \\ 0, & \text{if } \tau < 0 \end{cases}$$

$\mathcal{H}_2$ norm and Gramians

The squared of the $\mathcal{H}_2$ norm of a transfer matrix $G(s)$ can be computed as follows:

$$\|G(s)\|_2^2 = \int_0^\infty \mathrm{tr}(H(\tau)H^T(\tau))\mathrm{d}\tau = \int_0^\infty \mathrm{tr}\left(Ce^{A\tau}BB^Te^{A^T\tau}C^T\right)\mathrm{d}\tau$$

$$= \mathrm{tr}\left(C\int_0^\infty e^{A\tau}BB^Te^{A^T\tau}\mathrm{d}\tau\, C^T\right) = \mathrm{tr}(CW_cC^T)$$

where $W_c = W_c^T$ is the controllability Gramian, or as follows:

$$\|G(s)\|_2^2 = \int_0^\infty \mathrm{tr}(H^T(\tau)H(\tau))\mathrm{d}\tau = \int_0^\infty \mathrm{tr}\left(B^Te^{A^T\tau}C^TCe^{A\tau}B\right)\mathrm{d}\tau$$

$$= \mathrm{tr}\left(B^T\int_0^\infty e^{A^T\tau}C^TCe^{A\tau}\mathrm{d}\tau\, B\right) = \mathrm{tr}(B^TW_oB)$$

where $W_o = W_o^T$ is the observability Gramian.

$\mathcal{H}_2$ norm and Gramians

Matrices W_o and W_c are the solutions of the following Lyapunov equations:

$$A^T W_o + W_o A + C^T C = 0$$

$$A W_c + W_c A^T + B B^T = 0$$

To get the first Lyapunov equation, let's multiply the following

$$W_o = \int_0^\infty e^{A^T \tau} C^T C e^{A\tau} \mathrm{d}\tau$$

by $e^{A^T t_0}$ on the left and $e^{A t_0}$ on the right

$$e^{A^T t_0} W_o e^{A t_0} = \int_0^\infty e^{A^T(\tau + t_0)} C^T C e^{A(\tau + t_0)} \mathrm{d}\tau = \int_{t_0}^\infty e^{A^T \tau} C^T C e^{A\tau} \mathrm{d}\tau$$

Differentiating w.r.t. t_0 and assuming that matrix A is Hurwitz (i.e. the natural modes converge as $t \to \infty$), and that $g(x) = \int_{a(x)}^{b(x)} f(\tau)\mathrm{d}\tau \Longrightarrow g'(x) = f\big(b(x)\big)b'(x) - f\big(a(x)\big)a'(x)$, we have

$\mathcal{H}_2$ norm and Gramians

$$A^T e^{A^T t_0} W_o e^{A t_0} + e^{A^T t_0} W_o e^{A t_0} A = -e^{A^T t_0} C C^T e^{A t_0}$$

and finally setting $t_0 = 0$ leads to the following Lyapunov equation:

$$A^T W_o + W_o A + C^T C = 0$$

Remark: a similar result can be obtained for the controllability Gramian W_c.

Remark: In the *Matlab control toolbox* can be solved using the routine *lyap*.

The Hardy ∞-space $\mathcal{H}_\infty$

For a dynamic system (SISO and MIMO), the $\mathcal{H}_\infty$ norm provides a measure of the worst-case system gain.

If G is the transfer function matrix of a linear, time-invariant system, then G is a stable system *if and only if* $z = Gw \in \mathcal{H}_2$ whenever $w \in \mathcal{H}_2$ → L_2 is isomorphic to $\mathcal{H}_2$.
Since $z \in \mathcal{H}_2$ requires that z is analytic in the open-right-half-plane, a *necessary* condition for stability is that G is analytic in the open-right-half-plane.
A *sufficient* condition for $\|z\|_2 < \infty$ is that

$$\sup_{\alpha>0}\{\sup_\omega \bar{\sigma}(G(\alpha + j\omega))\} < \infty$$

since

$$\begin{aligned}\|\boldsymbol{G}w\|_2^2 &= \sup_{\alpha>0} \frac{1}{2\pi}\int_{-\infty}^{\infty} \|\boldsymbol{G}(\alpha+j\omega)w(\alpha+j\omega)\|^2\, d\omega \\ &\leq \sup_{\alpha>0} \frac{1}{2\pi}\int_{-\infty}^{\infty} \bar{\sigma}\big(\boldsymbol{G}(\alpha+j\omega)\big)^2 \|w(\alpha+j\omega)\|^2\, d\omega\end{aligned}$$

The Hardy ∞-space $\mathcal{H}_\infty$

$$\begin{aligned} &\leq \left\{\sup_{\alpha>0}\sup_{\omega}\bar{\sigma}\big(\boldsymbol{G}(\alpha+j\omega)\big)^2\right\}\sup_{\alpha>0}\frac{1}{2\pi}\int_{-\infty}^{\infty}\|w(\alpha+j\omega)\|^2\,d\omega \\ &= \sup_{\alpha>0}\sup_{\omega}\bar{\sigma}\big(\boldsymbol{G}(\alpha+j\omega)\big)^2\|w\|_2^2. \end{aligned}$$

The Hardy ∞-space $\mathcal{H}_\infty$ consists of functions of a complex variable that are analytic in the open-right-half-plane and such that the norm

$$\|G\|_\infty = \sup_{\alpha>0}\{\sup_\omega \bar{\sigma}(G(\alpha+j\omega))\}$$

is finite. That is

$$\mathcal{H}_\infty = \{G\colon G \text{ is analytic in } R_e(s) > 0 \text{ and } \|G\|_\infty < \infty\}$$

A transfer function matrix G defines a stable system if and only if $G \in \mathcal{H}_\infty$. If $G \in \mathcal{H}_\infty$ the limit $\mathrm{G_b(j\omega)} = \lim_{\alpha\to 0} G(\alpha+j\omega)$ exists for almost all ω. Furthermore, the mapping $G \longmapsto G_b$ is linear, injective and $\|G\|_\infty = \sup_\omega \bar{\sigma}(G(j\omega))$ ($\|G\|_\infty = \max_\omega \bar{\sigma}(G(j\omega))$).

$\mathcal{H}_\infty$ norm

In the case that G is rational, $G \in \mathcal{H}_\infty$ *if and only if* G has no pole in the closed-right-half-plane.

The $\mathcal{H}_\infty$-norm of a stable transfer matrix G is the induced norm defined by the maximum value of L_2 norms of its output vector signal z, assuming that the L_2 norms of its input vector signal w is 1:

$$\|G(s)\|_\infty = \max_{\|w(t)\|_2=1} \|z(t)\|_2 = \max_\omega \bar{\sigma}(G(j\omega))$$

This equation indicates that $\mathcal{H}_\infty$-norm is a *measure of the maximum gain* of a stable plant over all frequencies ω, which is equal to the energy gain of the system.

The terminology "$\mathcal{H}_\infty$ control theory" derives from the fact that we would like to achieve objective on $\|\cdot\|_\infty$ subject to closed-loop stability.

$\mathcal{H}_\infty$ norm and the Hamiltonian matrix

Unlike $\mathcal{H}_2$-norm , the $\mathcal{H}_\infty$-norm cannot usually be computed analytically. Only numerical solutions can be obtained.

1) Plot $\bar{\sigma}\big(G(j\omega)\big)$ as a function of ω and determine the maximum by inspection.

2) Search over ω by a computer. Such an approach cannot determine whether $\|G\|_\infty < \gamma$ because for any ω_0, the evaluation of $\bar{\sigma}\big(G(j\omega_0)\big)$ merely informs us that $\|G\|_\infty \geq \bar{\sigma}\big(G(j\omega_0)\big)$. If we have smoothness information about $\bar{\sigma}\big(G(j\omega)\big)$, we can determine $\|G\|_\infty$ from such an approach to any desired accuracy.

3) Alternatively, if we have a method to determine whether $\|G\|_\infty < \gamma$, then $\|G\|_\infty$ may be found from

$$\|G\|_\infty = \inf_\gamma \{\|G\|_\infty < \gamma\}$$

$\mathcal{H}_\infty$ norm and the Hamiltonian matrix

The last technique of finding $\|G\|_\infty$ involves a search over $\gamma > 0$. For each γ, we test whether $\|G\|_\infty < \gamma$ and then decrease or increase γ accordingly. The algorithm may be designed so that at any time we can stop and conclude that $\gamma_{low} < \|G\|_\infty < \gamma_{high}$.

As by Parseval's theorem, the $\mathcal{H}_\infty$ norm can be characterized in the time domain as

$$\|G\|_\infty = \sup\left\{\frac{\|z\|_2}{\|v\|_2} : \ v \neq 0\right\}$$

It follows that for any $\gamma > 0$, $\|G\|_\infty < \gamma$ if and only if

$$\begin{aligned} J_\infty(G,\gamma) &:= \max_v \left[\|z\|_2^2 - \gamma^2\|v\|_2^2\right] \\ &= \max_v \int_0^\infty \left[z^T(t)z(t) - \gamma^2 v^T(t)v(t)\right] dt < \infty \end{aligned}$$

The method provided in next slide is alternative and more practical to determine the maximum of $\bar{\sigma}\big(G(j\omega)\big)$.

$\mathcal{H}_\infty$ norm and the Hamiltonian matrix

The algorithm for the calculation of the $\mathcal{H}_\infty$-norm can be obtained by observing that $\|G(s)\|_\infty < \gamma$ (or $\bar{\sigma}\big(G(j\omega)\big) < \gamma$ for all ω (MIMO), $|G(j\omega)| < \gamma$ for all ω (SISO)) is equivalent to find condition on (A, B, C, D) of G (Doyle et. Al., 1989):

1. $\mathrm{Q}(\mathrm{j}\omega) = I - \frac{1}{\gamma^2} G^T(-j\omega) G(j\omega)$ is invertible for all $\omega \in \mathbb{R}$;
2. $Q(j\omega)^{-1}$ has no pole on the imaginary axis.

The next step is to construct a realization of $Q(j\omega)^{-1}$ and use the fact that the poles coincide with the eigenvalues of the corresponding Hamiltonian matrix H:

$$H = \begin{bmatrix} A & \frac{1}{\gamma^2} B B^T \\ -C^T C & -A^T \end{bmatrix}$$

$\mathcal{H}_\infty$ norm and the Hamiltonian matrix

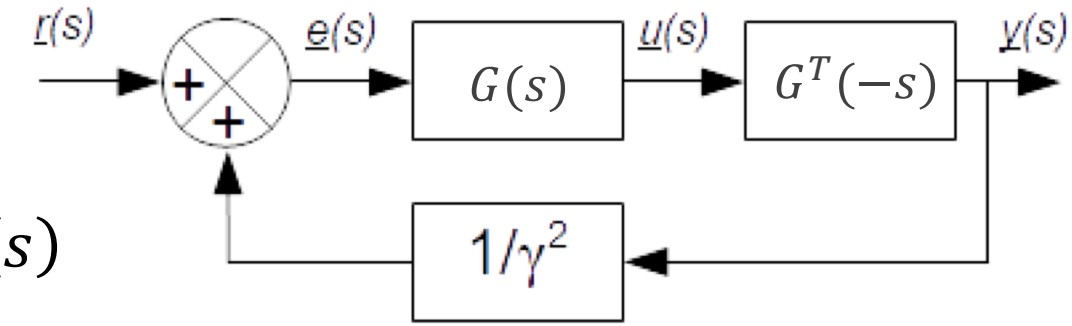
The key point is the achievement of the realization $\mathrm{Q}^{-1}(\mathrm{j}\omega)$ $= \left(I - \frac{1}{\gamma^2} G^T(-j\omega)G(j\omega)\right)^{-1}$. Let (A, B, C) be the realization of a strictly proper transfer matrix $G(s)$, then

$$\left(I - \frac{1}{\gamma^2} G^T(-j\omega)G(j\omega)\right)^{-1} = \left(\begin{array}{cc|c} \mathbf{A} & \frac{1}{\gamma^2}\mathbf{B}\mathbf{B}^T & \frac{\mathbf{B}}{\gamma} \\ -\mathbf{C}^T\mathbf{C} & -\mathbf{A}^T & \mathbf{0} \\ \hline \mathbf{0} & \frac{\mathbf{B}^T}{\gamma} & \mathbb{I} \end{array}\right)$$

$$:= \begin{bmatrix} 0 & \frac{B^T}{\gamma} \end{bmatrix} (sI - H)^{-1} \begin{bmatrix} \frac{B}{\gamma} \\ 0 \end{bmatrix} + I$$

$\mathcal{H}_\infty$ norm and the Hamiltonian matrix

The relation between $e(s)$ and $r(s)$ is:



$$e(s) = r(s) + \frac{1}{\gamma^2} G^T(-s) G(s) e(s)$$

$$\Longrightarrow e(s) = \left(I - \frac{1}{\gamma^2} G^T(-s) G(s) \right)^{-1} r(s)$$

On the other hand, the realization of $G^T(-s)$ is obtained from the realization of $G(s)$ as follows:

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & 0 \end{array} \right] := C(sI - A)^{-1} B \Rightarrow \begin{cases} \dot{x}_1 = A x_1 + B e \\ u = C x_1 \end{cases}$$

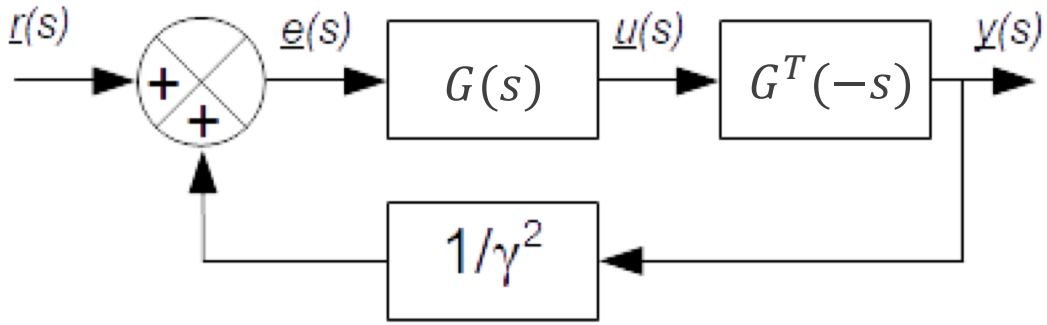
$$G^T(-s) = (C(-sI - A)^{-1} B)^T = -B^T(-sI - (-A^T))^{-1} C^T =$$

$$= \left[\begin{array}{c|c} -A^T & -C^T \\ \hline B^T & 0 \end{array} \right] \Rightarrow \begin{cases} \dot{x}_2 = -A^T x_2 - C^T u \\ y = B^T x_2 \end{cases}$$

$\mathcal{H}_\infty$ norm and the Hamiltonian matrix

We have

$$e = r + \frac{1}{\gamma^2} y$$



Thus, the realization of the feedback scheme in figure reads as follows:

$$\begin{cases} e = r + \dfrac{1}{\gamma^2} y \\ u = C x_1 \\ y = B^T x_2 \end{cases} \Rightarrow \begin{cases} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} A & \frac{1}{\gamma^2} B B^T \\ -C^T C & -A^T \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} r \\ y = \begin{bmatrix} 0 & B^T \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + I r \end{cases}$$

while in frequency domain we get

$$e(s) = \left(\begin{bmatrix} 0 & B^T \end{bmatrix} (sI - H)^{-1} \begin{bmatrix} B \\ 0 \end{bmatrix} + I \right) r(s)$$

$\mathcal{H}_\infty$ norm and the Hamiltonian matrix

Note also that:

$$\det\left(I - \frac{1}{\gamma^2} G^T(-s)G(s)\right) = 0 \Leftrightarrow \det\left(\gamma^2 I - G^T(-s)G(s)\right) = 0$$

and hence

$$\begin{aligned}
\gamma^2 I - G^T(-s)G(s) &= \gamma^2 I + B^T(sI - (-A^T))^{-1}C^T C(SI - A)^{-1}B \\
&= \gamma^2 I - B^T(-sI - A^T)^{-1}C^T C(sI - A)^{-1}B \\
&= \begin{bmatrix} (-sI - A)^{-1}B \\ 0 \end{bmatrix}^T \begin{bmatrix} -C^T C & 0 \\ 0 & \gamma^2 I \end{bmatrix} \begin{bmatrix} (sI - A)^{-1}B \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} (sI - A)^{-1}B \\ 0 \end{bmatrix}^* \begin{bmatrix} -C^T C & 0 \\ 0 & \gamma^2 I \end{bmatrix} \begin{bmatrix} (sI - A)^{-1}B \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} G(j\omega) \\ 0 \end{bmatrix}^* \begin{bmatrix} I & 0 \\ 0 & \gamma^2 I \end{bmatrix} \begin{bmatrix} G(j\omega) \\ 0 \end{bmatrix}
\end{aligned}$$

Kalman-Yakubovich-Popov lemma

Lemma (KYP): Given matrices A, B and the Hermitian matrix $\begin{bmatrix} P & S \\ S^T & R \end{bmatrix}$ the following two conditions are equivalent:

1. For all $\omega \in \mathbb{R}$, the frequency domain inequality $\Phi(\omega) = \Phi^T(\omega) > 0$ holds where:

$$\Phi(\omega) = \begin{bmatrix} (j\omega - A)^{-1}B \\ I \end{bmatrix}^* \begin{bmatrix} P & S \\ S^T & R \end{bmatrix} \begin{bmatrix} (j\omega - A)^{-1}B \\ I \end{bmatrix}$$

2. The following Linear Matrix Inequality (LMI) admits a Hermitian solution X (i.e. $X = X^*$):

$$\begin{bmatrix} A & B \\ I & 0 \end{bmatrix}^* \begin{bmatrix} 0 & X \\ X & 0 \end{bmatrix} \begin{bmatrix} A & B \\ I & 0 \end{bmatrix} + \begin{bmatrix} P & S \\ S^T & R \end{bmatrix} > 0$$

$\mathcal{H}_\infty$ norm and the Hamiltonian matrix

Schur complement: *Given a block matrix*

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

the Schur complement of D (A) *in* M *(assuming* D (A) *invertible) is*

$$M/D := A - BD^{-1}C \ (M/A := D - CA^{-1}B)$$

Consequently, previous LMI

$$\begin{bmatrix} A & B \\ I & 0 \end{bmatrix}^* \begin{bmatrix} 0 & X \\ X & 0 \end{bmatrix} \begin{bmatrix} A & B \\ I & 0 \end{bmatrix} + \begin{bmatrix} P & S \\ S^T & R \end{bmatrix} = \begin{bmatrix} XA + A^TX + P & XB + S \\ B^TX + S^T & R \end{bmatrix} > 0$$

is equivalent to the following inequality

$$XA + A^TX + P - (XB + S)R^{-1}(B^TX + S^T) > 0$$

that by replacing “>” with “=” is the generalized ARE in X.

$\mathcal{H}_\infty$ norm and the Hamiltonian matrix

Remark: with $\begin{bmatrix} P & S \\ S^T & R \end{bmatrix} = \begin{bmatrix} -C^TC & 0 \\ 0 & \gamma^2 I \end{bmatrix}$, we have as a special case of the KYP lemma, the Bounded real lemma and $\|G(s)\|_\infty < \gamma$ is equivalent to the fact that the Hamiltonian:

$$H = \begin{bmatrix} A & \frac{1}{\gamma^2}BB^T \\ -C^TC & -A^T \end{bmatrix}$$

has no pole on the imaginary axis. This in turn is equivalent to the following LMI

$$\begin{bmatrix} XA + A^TX - C^TC & XB \\ B^TX & \gamma^2 I \end{bmatrix} < 0$$

which is equivalent to the following inequality

$$XA + A^TX + C^TC - \frac{1}{\gamma^2}XBB^TX < 0$$

$\mathcal{H}_\infty$ norm and the bounded real lemma

Suppose $G = C(sI - A)^{-1}B + D$ with A asymptotically stable. The problem of finding the $\mathcal{H}_\infty$ norm of the transfer function matrix G is equivalent to the following optimization problem (LMI) in $P^T = P > 0$ (to be solved numerically in, e.g., Matlab©)

$$\min \gamma$$

$$\text{s.t.} \begin{bmatrix} A^T P + PA & PB & C^T \\ B^T P & -\gamma^2 I & D^T \\ C & D & I \end{bmatrix} < 0$$

Remark: by apply the *Schur complement* it is possible to show that previous minimization problem is equivalent the following lemma.

$\mathcal{H}_\infty$ norm and the bounded real lemma

Lemma: suppose $G = C(sI - A)^{-1}B + D$ with A asymptotically stable. Then $\|G\|_\infty < \gamma$ *if and only if*

1. $\|D\| < \gamma$ (equivalently $\bar{\sigma}(D) < \gamma$ or $R = \gamma^2 I - D^T D > 0$);
2. There exists $P = P^T$ satisfying the Riccati equation

$$P(A + BR^{-1}D^T C) + (A + BR^{-1}D^T C)^T P + \\ + PBR^{-1}B^T P + C^T(I + DR^{-1}D^T)C = 0 \quad (1)$$

such that $A + BR^{-1}(D^T C + B^T P)$ is asymptotically stable. Furthermore, if such a P exists, $\mathrm{P}^{\mathrm{T}} = P \geq 0$.

Remark: the "=" in (1) can be replaced by "<", but with $P > 0$.

$\mathcal{H}_\infty$ norm and the bounded real lemma

Theorem: suppose $G = C(sI - A)^{-1}B + D$ with A asymptotically stable. Then $\|G\|_\infty < \gamma$ *if and only if*

1. $\|D\| < \gamma$ (equivalently $\bar{\sigma}(D) < \gamma$ or $R = \gamma^2 I - D^T D > 0$);
2. The Hamiltonian

$$H = \begin{bmatrix} A + BR^{-1}D^T C & BR^{-1}B^T \\ -C^T(I + DR^{-1}D^T)C & -(A + BR^{-1}D^T C)^T \end{bmatrix}$$

with $R = \gamma^2 I - D^T D$, has no eigenvalues on the imaginary axis.

Therefore, the $\mathcal{H}_\infty$ norm is the smallest value of γ such that the Hamiltonian matrix H has no eigenvalues on the imaginary axis.

Remark: It is clear that $\|G\|_\infty < \gamma$ if and only if $\|\gamma^{-1}G\|_\infty < 1$. Hence there is no loss of generality to assume $\gamma = 1$. This assumption will often be made in the remainder of this course.

$\mathcal{H}_\infty$ norm and the bounded real lemma

Previous Lemmas suggest the following bisection algorithm to compute the $\mathcal{H}_\infty$ norm:

1. Select an upper bound γ_u and a lower bound γ_l such that $\gamma_l \leq \|G\|_\infty \leq \gamma_u$;
2. If $\frac{\gamma_u - \gamma_l}{\gamma_l} \leq$specified level, stop; $\|G\|_\infty \approx \frac{\gamma_u + \gamma_l}{2}$. Otherwise go to next step;
3. Set $\gamma = \frac{\gamma_u + \gamma_l}{2}$;
4. Test if $\|G\|_\infty < \gamma$ by calculating the eigenvalue of H for the given γ;
5. If H has an eigenvalue on $j\mathbb{R}$ set $\gamma_l = \gamma$; otherwise set $\gamma_u = \gamma$; go back to step 2.

$\mathcal{H}_\infty$ norm and the bounded real lemma

We may also use the Routh-Hurwitz stability criterion on the characteristic polynomial $\chi_H(s) := \det(sI - H)$ to get the $\mathcal{H}_\infty$ norm of $G(s)$.

Because $\chi_H(s)$ is an even polynomial involving powers of s^2, a row of zeros will appear in the Routh table of $\chi_H(s)$.
To build the Routh table, the row of zeros shall be replaced by the coefficients of the auxiliary polynomial obtained by taking the derivative with respect to s of the polynomial corresponding to the previous row.
Let $\{\gamma_i\}$ be the set of values of $\gamma \geq 0$ such that the γ –dependent coefficients in the first column of the Routh table becomes zero.
Then, the $\mathcal{H}_\infty$ norm of $G(s)$ is obtained as:

$$\|G(s)\|_\infty = \sup_i \{\gamma_i\}$$

$\mathcal{H}_\infty$ norm and the bounded real lemma

Example: Let $G(s)$ be defined as follows:

$$G(s) = \frac{1}{s+2}$$

To calculate the $\mathcal{H}_\infty$ norm of $G(s)$ we use the fact that:

$$G(j\omega) = \frac{1}{2+j\omega} \Rightarrow |G(j\omega)| = \frac{1}{\sqrt{4+\omega^2}}$$

and hence

$$\|G(s)\|_\infty = \sup_\omega |G(j\omega)| = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

A realization of $G(s)$ is $(A, B, C, D) = (-2,1,1,0)$. The Hamiltonian matrix is defined as follows:

$$H = \begin{bmatrix} A & \frac{1}{\gamma^2} BB^T \\ -C^T C & -A^T \end{bmatrix} = \begin{bmatrix} -2 & \frac{1}{\gamma^2} \\ -1 & 2 \end{bmatrix}$$

$\mathcal{H}_\infty$ norm and the bounded real lemma

The characteristic polynomial $\chi_H(s)$ of H reads:

$$\chi_H(s) = \det(sI - H) = \det\left(\begin{bmatrix} s+2 & -\dfrac{1}{\gamma^2} \\ 1 & s-2 \end{bmatrix}\right) = s^2 - 4 + \frac{1}{\gamma^2}$$

The Routh table corresponding to $\chi_H(s)$ is reported on the side. Note that coefficient 2 in row s_1 has been obtained by taking the derivative with respect to s of the polynomial corresponding to the previous row

s^2	1	$-4+\frac{1}{\gamma^2}$
s^1	$0 \to 2$	
s^0	$-4+\frac{1}{\gamma^2}$	

$$\frac{d}{ds}\left(s^2 - 4 + \frac{1}{\gamma^2}\right) = 2$$

The γ-dependent coefficient in the first column of the Routh table becomes zero for $\gamma \geq 0$ such that $\gamma = 1/2$: this is actually $\mathcal{H}_\infty$ norm of $G(s)$.

$\mathcal{H}_\infty$ norm and the bounded real lemma

Example: Let $G(s)$ be defined as follows:

$$G(s) = \begin{bmatrix} \frac{1}{s+1} & \frac{1}{s+2} \end{bmatrix}^T$$

To calculate the $\mathcal{H}_\infty$ norm of $G(s)$ we use the fact that:

$$G(j\omega) = \begin{bmatrix} \frac{1}{j\omega+1} \\ \frac{1}{j\omega+2} \end{bmatrix} \Rightarrow G^H(j\omega)G(j\omega) = \begin{bmatrix} \frac{1}{-j\omega+1} & \frac{1}{-j\omega+2} \end{bmatrix} \begin{bmatrix} \frac{1}{j\omega+1} \\ \frac{1}{j\omega+2} \end{bmatrix}$$

$$= \frac{1}{\omega^2+1} + \frac{1}{\omega^2+4}$$

Thus

$$\bar{\sigma}\big(G(j\omega)\big) = \sqrt{\frac{1}{\omega^2+1} + \frac{1}{\omega^2+4}}$$

$\mathcal{H}_\infty$ norm and the bounded real lemma

$$\|G(s)\|_\infty = \gamma = \sup_\omega \bar{\sigma}\big(G(j\omega)\big) = \sqrt{1+\frac{1}{4}} = \sqrt{\frac{5}{4}}$$

To compute the singular values of $G(j\omega)$, let's compute the characteristic polynomial of matrix $G(j\omega)G^H(j\omega)$:

$$\det(\lambda I - G(j\omega)G^H(j\omega)) = \frac{\lambda^2\omega^4 + 5\lambda^2\omega^2 + 4\lambda^2 - 2\lambda\,\omega^2 - 5\lambda}{(1+j\omega)(1-j\omega)(2+j\omega)(2-j\omega)}$$

$$= \frac{\lambda(\lambda(\omega^4 + 5\omega^2 + 4) - 2\omega^2 - 5)}{(1+j\omega)(1-j\omega)(2+j\omega)(2-j\omega)}$$

Thus, the eigenvalues of $G(j\omega)G^H(j\omega)$ are $\lambda = 0$ and $\lambda = \frac{2\omega^2+5}{\omega^4+5\omega^2+4}$.

The root square of the non-zero eigenvalue reads:

$$\sigma(G(j\omega) = \bar{\sigma}\big(G(j\omega)\big) = \sqrt{\frac{2\omega^2+5}{\omega^4+5\omega^2+4}} = \sqrt{\frac{1}{\omega^2+1} + \frac{1}{\omega^2+4}}$$

$\mathcal{H}_\infty$ norm and the bounded real lemma

Alternatively, a realization of $G(s)$ reads

$$G(s) = \frac{1}{s^2+3s+2}\begin{bmatrix} s+2 \\ s+1 \end{bmatrix} := \left[\begin{array}{c|c} A & B \\ \hline C & 0 \end{array}\right] = \left[\begin{array}{cc|c} 0 & 1 & 0 \\ -2 & -3 & 1 \\ \hline 2 & 1 & 0 \\ 1 & 1 & 0 \end{array}\right]$$

Then the Hamiltonian matrix H is defined as follows:

$$H = \begin{bmatrix} A & \frac{1}{\gamma^2}BB^T \\ -C^TC & -A^T \end{bmatrix} := \left[\begin{array}{cc|cc} 0 & 1 & 0 & 0 \\ -2 & -3 & 0 & \frac{1}{\gamma^2} \\ \hline -5 & -3 & 0 & 2 \\ -3 & -2 & -1 & 3 \end{array}\right]$$

The characteristic polynomial $\chi_H(s)$ of H reads:

$$\chi_H(s) = \det(sI - H) = \det\left(\begin{bmatrix} s & -1 & 0 & 0 \\ 2 & s+3 & 0 & -\frac{1}{\gamma^2} \\ 5 & 3 & s & -2 \\ 3 & 2 & 1 & s-3 \end{bmatrix}\right) = s^4 + \left(\frac{2}{\gamma^2} - 5\right)s^2 + 4 - \frac{5}{\gamma^2}$$

$\mathcal{H}_\infty$ norm and the bounded real lemma

The Routh table reads corresponding to $\chi_H(s)$ reads:

s^4	1	$\frac{2}{\gamma^2}-5$	$4-\frac{5}{\gamma^2}$
s^3	$0 \to 4$	$0 \to 2\left(\frac{2}{\gamma^2}-5\right)$	
s^2	$2\left(\frac{2}{\gamma^2}-5\right)$	$4\left(4-\frac{5}{\gamma^2}\right)$	
s^1	$\frac{88}{\gamma^2}-84$		
s^0	$4\left(4-\frac{5}{\gamma^2}\right)$		

Note that coefficients in row s^3 have been obtained by taking the derivative with respect to s of the polynomial corresponding to the previous row:

$$\frac{d}{ds}\left(s^4+\left(\frac{2}{\gamma^2}-5\right)s^2+4-\frac{5}{\gamma^2}\right)=4s^3+2\left(\frac{2}{\gamma^2}-5\right)s$$

Let $\{\gamma_i\}=\left\{\sqrt{\frac{2}{5}},\sqrt{\frac{88}{84}},\sqrt{\frac{5}{4}}\right\}$ be the set of values of $\gamma \geq 0$ such that the γ-dependent coefficients in the first column of the Routh table becomes zero. Then the $\mathcal{H}_\infty$ norm of $G(s)$ is obtained as:

$$\|G(s)\|_\infty = \sup_i\{\gamma_i\} = \sqrt{5/4}$$

$\mathcal{H}_\infty$ norm and the bounded real lemma

Example: consider a system $g = \frac{1}{s-a}$, with $a < 0$ and realization $(a, 1,1,0)$. The Riccati equation of the Bounded real lemma is

$$2ap + \frac{p^2}{\gamma^2} + 1 = 0$$

Which has solution $p = \gamma^2(-a \pm \sqrt{a^2 - \gamma^{-2}})$.

Now $a + \frac{p}{\gamma^2} = \pm\sqrt{a^2 - \gamma^{-2}}$ is either pure imaginary or real and therefore can be asymptotically stable only if real. In this case, the solution with the minus sign is asymptotically stable.

Thus, $\|g\|_\infty < \gamma$ if and only if $\gamma > \frac{1}{|a|}$.

$$\|g\|_\infty = \inf_\gamma [\gamma > \|g\|_\infty] = \frac{1}{|a|}$$

$\mathcal{H}_\infty$ norm and the bounded real lemma

The Hamiltonian is

$$H = \begin{bmatrix} a & -\gamma^{-2} \\ -1 & -a \end{bmatrix}$$

and has eigenvalues $\pm\sqrt{a^2 - \gamma^{-2}}$. Clearly H has no imaginary axis eigenvalues if and only if $\gamma > \frac{1}{|a|} = \|g\|_\infty$.

$\mathcal{H}_\infty$ vs $\mathcal{H}_2$ norm

$\mathcal{H}_\infty$: "push down the peak of largest singular value";

$\mathcal{H}_2$: "push down whole thing" (all singular value over all frequencies);

Example: let us consider again $G(s) = \frac{1}{s+a}$. The $\boldsymbol{\mathcal{H}_\infty}$ **norm** is

$$||G(s)||_\infty = \max_\omega |G(j\omega)| = \max_\omega \frac{1}{(\omega^2 + a^2)^{\frac{1}{2}}} = \frac{1}{a}$$

while the $\boldsymbol{\mathcal{H}_2}$ **norm** reads

$$\|G(s)\|_2 = \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} \underbrace{|G(j\omega)|^2}_{\frac{1}{\omega^2+a^2}} d\omega \right)^{\frac{1}{2}} = \left(\frac{1}{2\pi a} \left[\tan^{-1}\left(\frac{\omega}{a}\right) \right]_{-\infty}^{\infty} \right)^{\frac{1}{2}} = \sqrt{\frac{1}{2a}}$$

$\mathcal{H}_\infty$ vs $\mathcal{H}_2$ norm

Alternatively, consider the impulse response

$$g(t) = \mathcal{L}^{-1}\left(\frac{1}{s+a}\right) = e^{-at}, t \geq 0$$

To get

$$\|g(t)\|_2 = \sqrt{\int_0^\infty (e^{-at})^2 dt} = \sqrt{\frac{1}{2a}}$$

as expected from the Parseval's theorem.
There exists no general relationship between the $\mathcal{H}_2$ and $\mathcal{H}_\infty$ norms (apart from special cases, see next slide)

$$f_1(s) = \frac{1}{\epsilon s + 1}, \quad f_2(s) = \frac{\epsilon s}{s^2 + \epsilon s + 1}$$

$\epsilon \longrightarrow 0$. We have

$$||f_1||_\infty = 1 \quad ||f_1||_2 = \infty$$

$$||f_2||_\infty = 1 \quad ||f_2||_2 = 0$$

$\mathcal{H}_\infty$ vs $\mathcal{H}_2$ norm

Let $G(s)$ be a strictly proper, stable transfer function of the following form with poles λ_i of multiplicity 1:

$$G(s) = \sum_i \frac{N_i}{s - \lambda_i}$$

Let $\lambda_i = -a_i + jb_i$ with $a_i > 0$. Then it can be shown that:

$$\|G(s)\|_\infty^2 \leq \|G(s)\|_2^2 \sum_i \frac{2}{a_i}$$

Where, by using the residue theorem, we have

$$\|G(s)\|_2^2 = \mathrm{tr}\left(\sum_i N_i G^T(-\lambda_i)\right)$$

$\mathcal{H}_\infty$ vs $\mathcal{H}_2$ norm

Why is the $\mathcal{H}_\infty$ norm so popular? It is convenient for modeling unstructured uncertainty, and because it satisfies the multiplicative property:

$$\|G_1(s)G_2(s)\|_\infty \leq \|G_1(s)\|_\infty \cdot \|G_2(s)\|_\infty$$

What is wrong with the $\mathcal{H}_2$ norm? It is *not* an induced norm, and it does *not* satisfy the multiplicative property:

$$\|G_1(s)G_2(s)\|_2 \nleq \|G_1(s)\|_2 \cdot \|G_2(s)\|_2$$

Example: Let $G(s) = 1/(s+a)$ for which $\|G(s)\|_2 = \sqrt{1/2a}$.

$$\|G(s)G(s)\|_2 = \sqrt{\int_0^\infty |\underbrace{\mathcal{L}^{-1}\left[\left(\frac{1}{s+a}\right)^2\right]}_{te^{-at}}|^2} = \sqrt{\frac{1}{a}\frac{1}{2a}} = \sqrt{\frac{1}{a}}\,\|G(s)\|_2^2$$

For $a < 1$, $\|G(s)G(s)\|_2 > \|G(s)\|_2 \cdot \|G(s)\|_2$

For the $\mathcal{H}_\infty$ norm: $\|G(s)G(s)\|_\infty = \dfrac{1}{a^2} = \|G(s)\|_\infty \cdot \|G(s)\|_\infty$

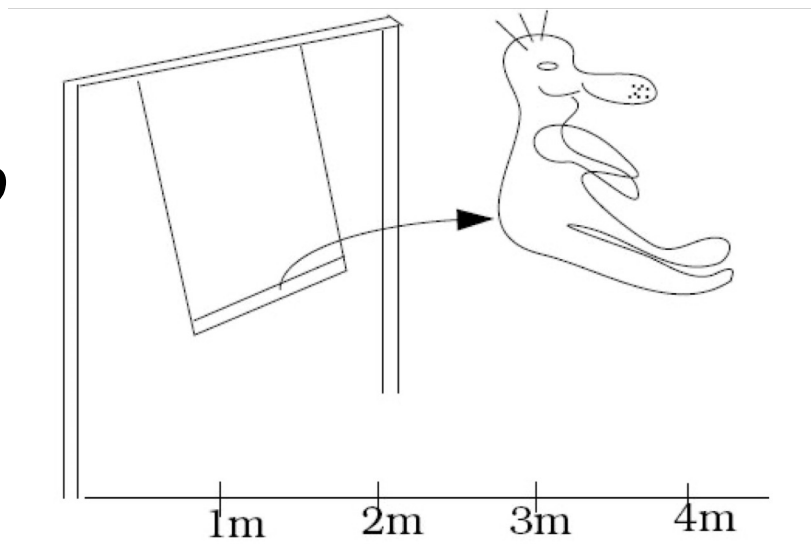
Hankel norm

$G(s)$ stable.

The Hankel norm is obtained when one applies an input $w(t)$ up to $t = 0$ and measures the output $z(t)$ for $t > 0$, and select an input $w(t)$ to maximise the ratio of the 2-norm of these two signals:

$$\|G(s)\|_H \triangleq \max_{w(t)} \frac{\sqrt{\int_0^\infty \|z(\tau)\|_2^2 d\tau}}{\sqrt{\int_{-\infty}^0 \|w(\tau)\|_2^2 d\tau}}$$

Its definition is analogous to trying to pump a swing with limited input energy such that the subsequent length of jump is maximized as illustrated in Figure.



Hankel norm

It may be show that the Hankel norm is equal to:

$$\|G(s)\|_H = \sqrt{\rho(PQ)}$$

where ρ is the spectral radius (absolute value of maximum eigenvalue), P is the controllability Gramian and Q the observability Gramian.

The name "Hankel" is used because the matrix PQ has the special structure of a Hankel matrix: which has identical elements along the "wrong-way" diagonals.

$$\sigma_i = \sqrt{\lambda_i(PQ)}$$ Hankel singular values

$$\|G(s)\|_H \equiv \sigma_1 \leq \|G(s)\|_\infty \leq 2\sum_{i=1}^{n} \sigma_i$$

$$A = \begin{bmatrix} a_0 & a_1 & a_2 & \dots & \dots & a_{n-1} \\ a_1 & a_2 & & & & \vdots \\ a_2 & & & & & \vdots \\ \vdots & & & & & a_{2n-4} \\ \vdots & & & & a_{2n-4} & a_{2n-3} \\ a_{n-1} & \dots & \dots & a_{2n-4} & a_{2n-3} & a_{2n-2} \end{bmatrix}.$$

Example

Example 4.22 *We want to compute analytically the various system norms for* $G(s) = 1/(s+a)$ *using state-space methods. A state-space realization is* $A = -a$, $B = 1$, $C = 1$ *and* $D = 0$. *The controllability Gramian* P *is obtained from the Lyapunov equation* $AP + PA^T = -BB^T \Leftrightarrow -aP - aP = -1$, *so* $P = 1/2a$. *Similarly, the observability Gramian is* $Q = 1/2a$.
$\mathcal{H}_2$ *norm is then*

$$\|G(s)\|_2 = \sqrt{\mathrm{tr}(B^T Q B)} = \sqrt{1/2a}$$

The eigenvalues of the Hamiltonian matrix H *in* BRL *are*

$$\lambda(H) = \lambda\begin{bmatrix} -a & 1/\gamma^2 \\ -1 & a \end{bmatrix} = \pm\sqrt{a^2 - 1/\gamma^2}$$

We £nd that H *has no imaginary eigenvalues for* $\gamma > 1/a$, *so*

$$\|G(s)\|_\infty = 1/a$$

The Hankel matrix is $PQ = 1/4a^2$ *and the Hankel norm is*

$$\|G(s)\|_H = \sqrt{\rho(PQ)} = 1/2a$$